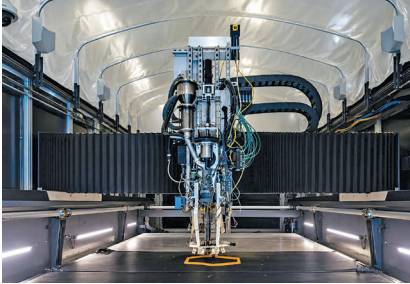


Smart monitoring 3D printers fix their own mistakes

Developed at Oak Ridge National Laboratory in Tennessee, US, the system maintains optimal temperatures for stronger layer bonding and structural consistency. A



A test object is 3D-printed using a new system to monitor for errors and correct them automatically while manufacturing large items made from plastic composite (Credit: Carlos Jones/ORNL, US Dept. of Energy)

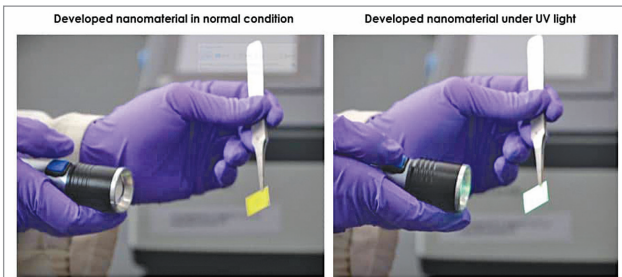
new AI-powered monitoring system can automatically detect and correct errors during large-scale 3D printing, improving reliability and reducing material waste.

The technology uses thermal imaging, computer vision, sensors, and machine learning

to monitor printing conditions in real time and adjust print speed when defects are detected. Tests on a large composite structure showed the platform could identify temperature deviations and autonomously correct them. Researchers also integrated a digital twin, enabling virtual testing of designs and manufacturing parameters.

Smart material developed to fight counterfeiting

Engineered by a team at the Indian Institute of Technology Guwahati, the technology uses perovskite nanocrystals with a double-layer protective coating that improves resistance to



IIT Guwahati develops next-generation security technology to combat fake products and forged documents (Credit: www.iitg.ac.in)

heat and chemicals while maintaining optical performance. Researchers have developed a smart light-emitting nanomaterial that could significantly strengthen protection against counterfeit currency, forged documents, and fake products by creating security patterns that are extremely difficult to replicate. The material enables microscopic, high-resolution patterns through direct laser writing, supporting secure authentication and data storage. Dubbed '4D anti-counterfeiting,' the approach could also aid development of advanced micro-LED displays and augmented reality devices.

Robots learn real-world tasks through simulation

Nvidia researchers have demonstrated multiple robotics frameworks that improve the performance of robots trained in simulation in real-world environments, helping to narrow the long-standing sim-to-real gap. Presented at the International Conference on Robotics and Automation (ICRA), the technologies enhance navigation, grasping, object manipulation, and decision-making. Engineered by teams at Nvidia in collaboration with researchers from Carnegie Mellon University, the University of Utah, and the University of Sydney, the systems achieved significant gains in real-world testing.

Quantum Li-Fi secures networks

Research conducted in Germany by a consortium led by KEEQuant and supported by Fraunhofer IPMS highlights a pathway toward flexible, high-security next-generation wire-



Close-up of the Li-Fi connection between the two QKD nodes (Credit: KeeQuant)

less networks. Quantum-secure Li-Fi technology has been shown to strengthen wireless network security by combining Li-Fi data transmission with quantum key distribution (QKD), enabling encrypted communication and detecting eavesdropping attempts.

The system supports secure optical wireless links without relying on conventional radio-frequency networks, offering potential applications in defence, critical infrastructure, and enterprise communications. Developed through the QuINSiDa project, the technology integrates quantum cryptography, key management, and Li-Fi networking into a unified platform.

Light-based computer chip saves power

Engineered by a team at the University of Florida in the United States, the silicon photonic chip combines optical and electronic components to execute AI computations faster and with significantly lower power consumption. The newly developed light-based computer chip could reduce the energy required for artificial intelligence workloads by up to 100 times compared with conventional electronic processors. The chip performs convolution operations, core calculations used in image recognition, and pattern detection using lasers and microscopic lenses instead of relying solely on electricity. Tests showed it achieved about 98% accuracy in classifying handwritten digits, matching the performance of traditional chips. The innovation could help address the growing electricity demands of AI systems.